Numerical Weather Prediction (NWP)

Individual Project

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What is NWP?

- **Definition:** Computer-based numerical simulation and prediction of atmospheric and oceanic systems
- One of the major scientific achievements of the past 50 years
- Core component of operational weather forecasting
- Enables accurate forecasts exceeding one week
- **Key Dependencies:**
  - Higher model resolution
  - Quality numerical schemes
  - Improved physical parameterizations
  - Satellite and observational data
  - Refined initial conditions

Fundamental Framework

Core Foundation

NWP is based on a system of coupled nonlinear partial differential equations (PDEs) derived from fundamental physics laws:

- Newton's laws of motion
- Conservation of energy
- Conservation of mass
- Conservation of water vapor
- Thermodynamic relationships

These equations describe the evolution of atmospheric state variables: velocity, temperature, pressure, density, and humidity.

Equation 1: Navier-Stokes Equations

Equation of Motion (Vector Form)

$$\frac{\partial \boldsymbol{V}}{\partial t} + \boldsymbol{V} \cdot \nabla \boldsymbol{V} + 2\boldsymbol{\Omega} \times \boldsymbol{V} + \frac{1}{\rho}\nabla p = \boldsymbol{F} + \boldsymbol{g}$$

Physical Interpretation:

- $\frac{\partial \boldsymbol{V}}{\partial t}$: Local acceleration
- $\boldsymbol{V} \cdot \nabla \boldsymbol{V}$: Nonlinear advection term
- $2\boldsymbol{\Omega} \times \boldsymbol{V}$: Coriolis effect ($\boldsymbol{\Omega}$ = Earth's rotation)
- $\frac{1}{\rho}\nabla p$: Pressure gradient force
- $\boldsymbol{F}$: Friction (important in boundary layer)
- $\boldsymbol{g}$: Apparent gravity (includes centrifugal force)

Equation 2: Equation of State

Ideal Gas Law for Atmosphere

$$p = R\rho T$$

Variables:

- p: Atmospheric pressure
- ρ: Air density
- T: Temperature
- R: Gas constant for dry air

Note: In practice, this equation must be corrected for atmospheric moisture content to account for water vapor effects.

Equation 3: Energy Conservation

First Law of Thermodynamics

$$c_v \frac{dT}{dt} + RT\nabla \cdot \boldsymbol{V} = Q$$

Physical Interpretation:

- c_v: Specific heat at constant volume
- $\frac{dT}{dt}$: Lagrangian time derivative of temperature
- $RT\nabla \cdot \boldsymbol{V}$: Work done by compression/expansion
- Q: Diabatic heating rate (radiation, latent heat, etc.)

This equation governs temperature evolution through advection, adiabatic processes, and external heating sources.

Equation 4: Mass Conservation

Continuity Equation

$$\frac{d\rho}{dt} + \rho\nabla \cdot \boldsymbol{V} = 0$$

Physical Meaning:

- Ensures conservation of mass in the atmospheric system
- $\frac{d\rho}{dt}$: Rate of density change following air parcel
- $\rho\nabla \cdot \boldsymbol{V}$: Density change due to divergence/convergence

Implication: Air density changes only through compression or expansion, not through creation or destruction of mass.

Equation 5: Water Vapor Conservation

Moisture Equation

$$\frac{dq}{dt} = S$$

Components:

- q: Specific humidity (mass of water vapor per unit mass of air)
- $\frac{dq}{dt}$: Lagrangian rate of change of humidity
- S: Net sources and sinks of water vapor

Sources/Sinks (S):

- Evaporation from surface ($+$)
- Condensation to form clouds ($-$)
- Precipitation ($-$)

Equation 6: Hydrostatic Balance

Hydrostatic Equation

$$\frac{\partial p}{\partial z} + g\rho = 0$$

Physical Interpretation:

- Valid for large-scale atmospheric motions
- Vertical pressure gradient balances gravitational force
- z: Vertical height coordinate
- Assumes vertical acceleration is negligible

Application:

- Early NWP models were all hydrostatic
- Modern high-resolution models often use non-hydrostatic formulations

Equation 7: Barotropic Vorticity Equation

Historical First NWP Equation (ENIAC, 1950)

$$\frac{d\zeta}{dt} = 0$$

where $\frac{d}{dt} = \frac{\partial}{\partial t} + \boldsymbol{V} \cdot \nabla$ and ζ is vorticity

Physical Meaning:

- Conservation of absolute vorticity following air parcels
- ζ: Relative vorticity of the flow field
- Simplified model used in the first successful computer forecast

Historical Significance: This simplified equation enabled the first successful 24-hour NWP forecast on ENIAC in 1950.

Initial Conditions: The Critical Challenge

Initial Value Problem

NWP solves an initial value problem: given atmospheric state at time $t = 0$, predict future states.

Historical Lesson:

- Richardson's 1922 forecast failed primarily due to inaccurate initial conditions
- Initial data was unbalanced, leading to unrealistic results

Modern Solution: Data Assimilation

- Combines background field (forecast) with observations
- Key methods: Optimal Interpolation, 3D-Var, 4D-Var
- Satellite data assimilation revolutionized NWP

Equation 8: 3D-Var Cost Function

Three-Dimensional Variational Assimilation

$$J = \frac{1}{2}(\boldsymbol{X} - \boldsymbol{X}_B)^T \boldsymbol{B}^{-1}(\boldsymbol{X} - \boldsymbol{X}_B) + \frac{1}{2}(\boldsymbol{Y} - H\boldsymbol{X})^T \boldsymbol{R}^{-1}(\boldsymbol{Y} - H\boldsymbol{X})$$

Components:

- $\boldsymbol{X}$: Model state vector (to be optimized)
- $\boldsymbol{X}_B$: Background field (first guess from forecast)
- $\boldsymbol{B}$: Background error covariance matrix
- $\boldsymbol{Y}$: Observation vector
- H: Observation operator (model $\rightarrow$ observations)
- $\boldsymbol{R}$: Observation error covariance matrix

Goal: Minimize J to find optimal initial state balancing background and observations.

4D-Var: Time-Distributed Assimilation

Extension of 3D-Var:

- Considers observations over a time window (e.g., 6-12 hours)
- Uses model dynamics to propagate information temporally
- Requires tangent linear model and adjoint model

Advantages over 3D-Var:

- Better utilizes asynchronous observations
- Implicitly balances initial conditions through model dynamics
- Higher forecast accuracy but computationally expensive

Breakthrough: Direct assimilation of satellite radiance data dramatically improved forecast skill, especially in Southern Hemisphere.

Numerical Discretization

Challenge

Continuous PDEs must be converted to discrete algebraic systems suitable for computers.

Key Methods:

1. **Time Integration:**
   - Leapfrog scheme (explicit, requires CFL stability)
   - Implicit schemes (unconditionally stable for linear terms)
2. **Spatial Discretization:**
   - Finite difference methods
   - Finite element methods
   - Spectral methods (used in global models)

Equation 9: Wave Equation Discretization

One-Dimensional Advection Equation

$$\frac{\partial Q}{\partial t} + c\frac{\partial Q}{\partial x} = 0$$

Centered Difference Discretization

$$\frac{Q_m^{n+1} - Q_m^{n-1}}{2\Delta t} + c\left(\frac{Q_{m+1}^n - Q_{m-1}^n}{2\Delta x}\right) = 0$$

Notation:

- $Q_m^n = Q(m\Delta x, n\Delta t)$: Value at grid point m, time step n
- Δt: Time step, Δx: Grid spacing
- c: Advection speed

Stability: CFL Criterion

Courant–Friedrichs–Lewy (CFL) Condition

For numerical stability of explicit schemes:

$$\frac{c\Delta t}{\Delta x} \leq 1$$

Physical Interpretation:

- Information should not travel more than one grid cell per time step
- Violated when time step too large or grid spacing too small
- Violation leads to non-physical exponential growth

Implication: Higher spatial resolution requires smaller time steps, increasing computational cost dramatically.

Equation 10: Spectral Method

Spherical Harmonic Expansion

$$Q(\lambda, \phi, t) = \sum_{n=0}^{N} \sum_{m=-n}^{n} Q_n^m(t) Y_n^m(\lambda, \phi)$$

Components:

- λ: Longitude, ϕ: Latitude
- $Y_n^m(\lambda, \phi)$: Spherical harmonic functions (basis functions)
- $Q_n^m(t)$: Time-dependent spectral coefficients
- N: Truncation wavenumber (determines resolution)

Advantages:

- High accuracy for smooth fields
- Efficient computation of derivatives
- Used in ECMWF's IFS global model

Model Types and Applications

1. **Global Models** (e.g., ECMWF IFS)
   - Medium-range forecasts (3-10 days)
   - Spectral methods
   - Resolution: T1279 ($\sim$16 km)
2. **Mesoscale Models** (e.g., WRF)
   - Short-range, high-resolution forecasts
   - Regional domains with flexible configuration
   - Nested grids for local refinement
3. **Ensemble Forecasting**
   - Multiple forecasts with perturbed initial conditions
   - Probabilistic predictions
   - Addresses atmospheric chaos and predictability limits

Predictability and Chaos

Fundamental Limitation:

- Atmosphere is a chaotic system
- Small errors in initial conditions grow exponentially
- Limits deterministic forecast skill to $\sim$2 weeks

Historical Progress:

- Forecast skill improves $\sim$1 day per decade
- Current 10-day forecasts as accurate as 7-day forecasts 30 years ago
- Northern and Southern Hemisphere skill now comparable (thanks to satellites)

Verification Metric:

- Anomaly Correlation Coefficient (ACC) $> 60\%$ indicates skillful forecast

Current Challenges

1. **Nowcasting (0-6 hours):**
   - Requires rapid multi-source data fusion
   - High-resolution radar and satellite integration
2. **Extended-Range Forecasts:**
   - Coupled atmosphere-ocean-sea ice modeling
   - Accurate representation of aerosols and trace gases
3. **Climate Prediction:**
   - Climate change modeling remains a major scientific challenge
   - Requires multi-decadal simulations with coupled Earth system models
4. **Physical Parameterizations:**
   - Sub-grid scale processes (clouds, convection, turbulence)
   - Major source of model uncertainty

Applications and Impact

Core Application: Extreme weather warnings

- Hurricanes, severe storms, floods
- Saves lives and reduces property damage

Broader Applications:

- **Transportation:** Aviation routing, marine navigation
- **Energy:** Wind/solar power forecasting, demand prediction
- **Agriculture:** Crop planning, frost warnings
- **Environmental:** Pollution dispersion modeling
- **Military:** Strategic planning
- **Economy:** Weather derivatives, risk management

Societal Value: NWP is fundamental infrastructure for modern society.

Summary

1. NWP is built on fundamental physics laws expressed as coupled nonlinear PDEs
2. Success requires interdisciplinary integration:
   - Physics
   - Mathematics
   - Computer Science
   - Atmospheric Science
3. Initial condition accuracy is paramount (data assimilation)
4. Numerical stability and accuracy are critical (CFL criterion)
5. Atmospheric chaos limits deterministic predictability
6. Continuous improvement: better resolution, physics, data, and methods